HDOC - 64th day

FINDING THE DUPLICATE ELEMENT USING POINTERS

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Problem Statement

Write a C program that reads the size and elements of a variable-length integer array, nums[]. The array contains exactly one element repeated once. Pass the array to a function named duplicate_element() using its address. Use pointers to pass the array between the calling function and duplicate_element(). The function duplicate_element() must:

- Find the only repeated element in nums[] and display the repeated element.
- Find the result in a single iteration through the array by using an efficient approach such as the XOR technique.
- Assume that every other element appears exactly once.

Algorithm

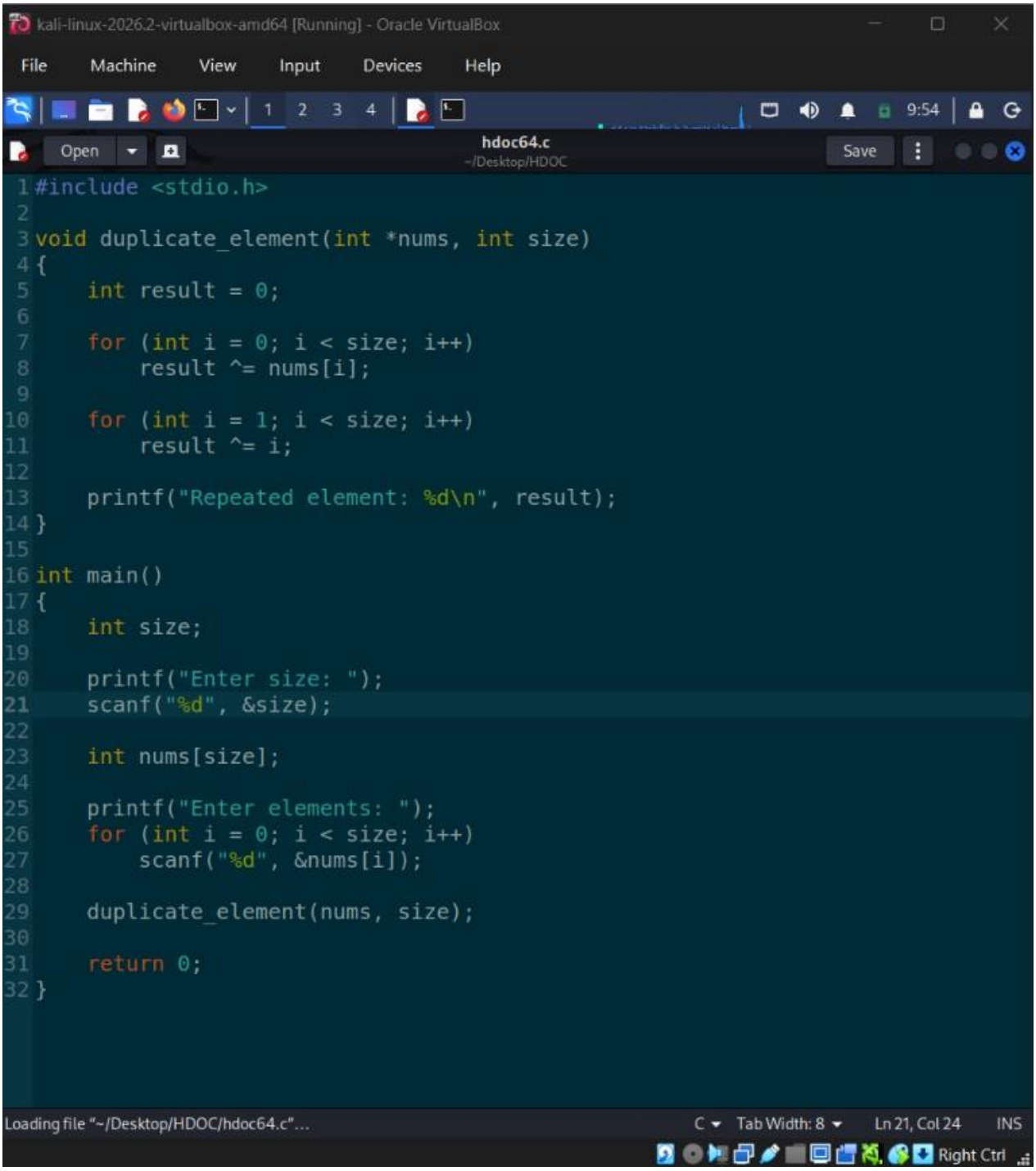
1. Start.
2. Read the size of the array nums[].
3. Read the elements of nums[].
4. Pass nums and size to duplicate_element() using the array address.
5. Initialize result = 0.

6. Traverse the array and XOR every element with result.
7. XOR result with all integers from 1 to size - 1.
8. The remaining value in result is the repeated element.
9. Display the repeated element.
10. Stop.

Test Case Table

S. No.	Size	nums[]	Expected Output	Actual Output	Result
1	5	1 2 3 4 3	3	3	Pass
2	6	1 2 3 4 5 2	2	2	Pass
3	4	1 2 3 1	1	1	Pass
4	3	1 2 2	2	2	Pass
5	7	1 2 3 4 5 6 5	5	5	Pass

C Program



Compilation and Execution

Compilation command: gcc hdoc64.c -o hdoc64

```
┌──(kali㉿kali)-[~/Desktop/HDOC]
└─$ gcc hdoc64.c -o hdoc64
```

Execution command: ./hdoc64 Execution

Screenshot:

